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From M Passes to One

A MapReduce-Style Bootstrap for Pointwise Off-Policy Evaluation in Production Recommenders

Ghulam Ahmed Ansari · Cheng Ding

LinkedIn Corporation

01 / MAKE FULL-SLATE OPE USEFUL ON FEED

From dedicated random sessions to production logs

Position-one replay baseline

Uniform-random logging (WSDM 2011).

→

Production logs + full-slate OPE

Deployed; dedicated random-session branch retired.

Requires logging support and propensity assumptions.

02 / THE ONE-PASS TILED CONSTRUCTION

Make uncertainty practical: 50 passes become one

M separate reads

Read 1 → estimate 1 | Read 2 → estimate 2 | ... | Read M → estimate M

ONE SHARED READ

Session's OPE contribution g(s)

session 1
session 2
session 3

Computed once

Same rows in every tile

SEEDED TILES (M = 50 REPLICAS)

	tile 1 mask slots	tile 2 mask slots	...	tile M mask slots
session 1	1 0 0 0	0 0 0 0		1 1 0 0
session 2	0 1 0 1	0 0 0 1		0 0 0 0
session 3	0 0 0 0	0 1 0 1		0 0 1 0
	weight × g(s)	weight × g(s)		weight × g(s)

REDUCE TO ESTIMATES

50 masked accumulators

Keep replicas separate; sum, then normalize

M estimates

↓

Sample variance

0 1 0 1 → session weight 2 (illustrative).

Within 1% relative variance error vs. same-mask multi-pass reference.

Masks and updates still scale with M.

No measured 50x speedup claim.

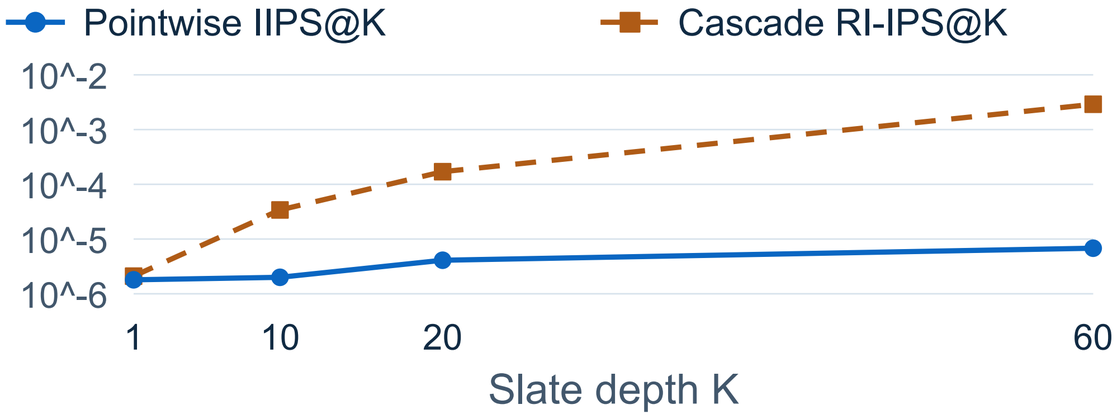
03 / TEST THE SLATE ASSUMPTION

Low pointwise error through K = 60

Pointwise assumes no preceding-item dependence.

Feed composite engagement; on-policy reference (Table 1).

Normalized mean squared error (log; lower is better)



Pointwise has lower error here; cause not isolated.

Unstable sequential weights are one possible reason.

04 / ALIGN SCORES ACROSS RANKER VERSIONS

Calibrate scores before computing propensities

Re-weighting can change score scales; raw softmax can confuse scale with preference.

Raw model scores → softmax: scale-sensitive

Equal-population score bins → **Observed response rates** → **Softmax**

Calibrate candidate and logging rankers separately.

Weight: candidate / logging probability (πe / πl).

Online-direction agreement: 1/2 → 2/2

Three ranker versions, two consecutive comparisons.

Calibration restored agreement after re-weighting.

05 / VALIDATE AGAINST ONLINE OUTCOMES

Does the complete system agree with online outcomes?

Second-pass ranker: same 11 model comparisons

Position-one replay baseline — **18% (2/11); 95% CI 5-48%**

Bootstrap IIPS@K — **64% (7/11); 95% CI 35-85%**

Lines: Wilson 95% confidence intervals.

Combined offline-online agreement (two surfaces)

70% (16/23) 95% CI 49-84%

Suggested Post: 75% (9/12), separate model set.

Agreement: same online direction and significance.

Kendall rank correlation +0.29; p=0.04; broader 26 pairs.

Comparison limits

Logging data and evaluated positions differ.

Small model sets; logging support assumed.

Remaining limitations

Earlier candidate-selection bias is not corrected.

Overall agreement is not attributed to each component.

Conference listing

programs.sigchi.org/recsys/2026/program/content/270632

Paper DOI: 10.1145/3773078.3831911

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